

LAB 1 Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
FR-001	Functional	The system shall track grant fund allocations, allow faculty to log equipment procurement expenses, and compute remaining budgetary balance.	High	Pass: Balance updates accurately and blocks expenses exceeding grant cap; Fall: Overdraft allowed without admin override.	Prevents grant overspending and maintains accurate financial records for sponsored research.
FR-002	Functional	The system shall allow faculty members to record research publication details including journal indexing status (Scopus/Web of Science) and DOI metadata.	High	Pass: System verifies indexing metadata against DOI database and logs entry on dashboard; Fall: Missing mandatory fields allows unverified record submission.	Essential for tracking institutional research output and academic indexing standards.
FR-003	Functional	The system shall execute an automated co-author approval workflow before finalizing publication submissions.	High	Pass: Status remains 'Pending' until all listed internal co-authors digitally sign approval; Fall: Submission finalized without explicit co-author confirmation.	Ensures academic integrity and verifies co-authorship consent prior to institutional logging.
FR-004	Functional	The system shall provide the Research Dean with real-time fund burn-up analytics and grant utilization visual dashboards.	Medium	Pass: Dashboards display real-time expenditure charts matching backend ledger totals within 2 seconds of log update; Fall: Discrepancy between analytics	Enables deanery leadership to monitor research expenditure trends and resource allocation.

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				visuals and backend logs.	
FR-005	Functional	The system shall automatically aggregate external citation metrics and calculate updated author h-index values on a monthly cycle.	Medium	Pass: System fetches updated citation counts via API and updates metrics without manual input; Fall: Citation count fails to sync or calculation error occurs.	Tracks institutional research impact and citation metrics required by the deanery.
NFR-001	Non-Functional	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals and journal status modifications.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load; Fall: Audit logs can be modified or deleted by non-root users.	Guarantees financial accountability, auditability, and data integrity for university grants.
NFR-002	Non-Functional	The system shall enforce Role-Based Access Control (RBAC) ensuring faculty can only edit their own grants while the Research Dean maintains institution-wide view permissions.	High	Pass: Faculty access is strictly restricted to assigned projects; Fall: Faculty user accesses or modifies another project's financial ledger.	Ensures privacy, regulatory compliance, and security of sensitive institutional data.